SpotCheck

Bridging the 'Reel vs. Real' Gap in Modern Travel

Authentic, real-time travel intelligence sourced directly from verified locals within your vicinity.

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Platform: Android (Jetpack Compose) + FastAPI Cloud Architecture
Product by : Thinaira Tech

APP OVERVIEW & VALUE PROPOSITION

Solving Ground Reality in Travel Planning

SpotCheck

⚠ The 'Reel' Trap

Travelers plan trips based on viral social reels and static star reviews that reflect ideal conditions from months ago, leaving them blind to sudden realities.

Dynamic Ground Reality

Sudden weather downpours, hill-road landslides, holiday crowd spikes, and unexpected temple queues change hourly.

The SpotCheck Solution

A hyper-local network connecting travelers directly with verified residents living within 20–25 km of the destination for live, real-time ground truth.

Key Guarantee

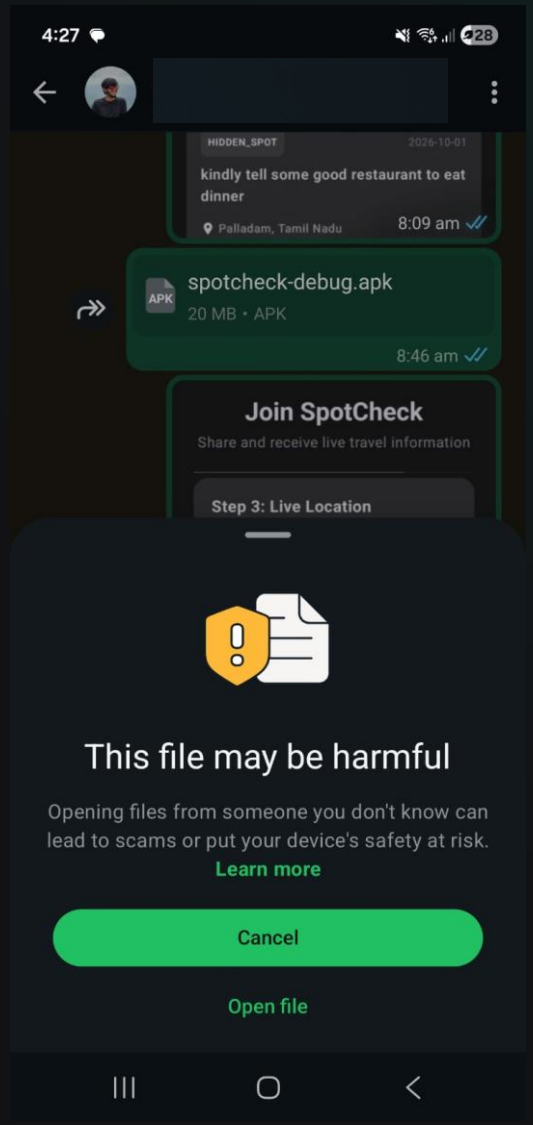
Zero stale information. Zero downloaded gallery pictures. Direct on-site visual proof and meter-accurate GPS telemetry.

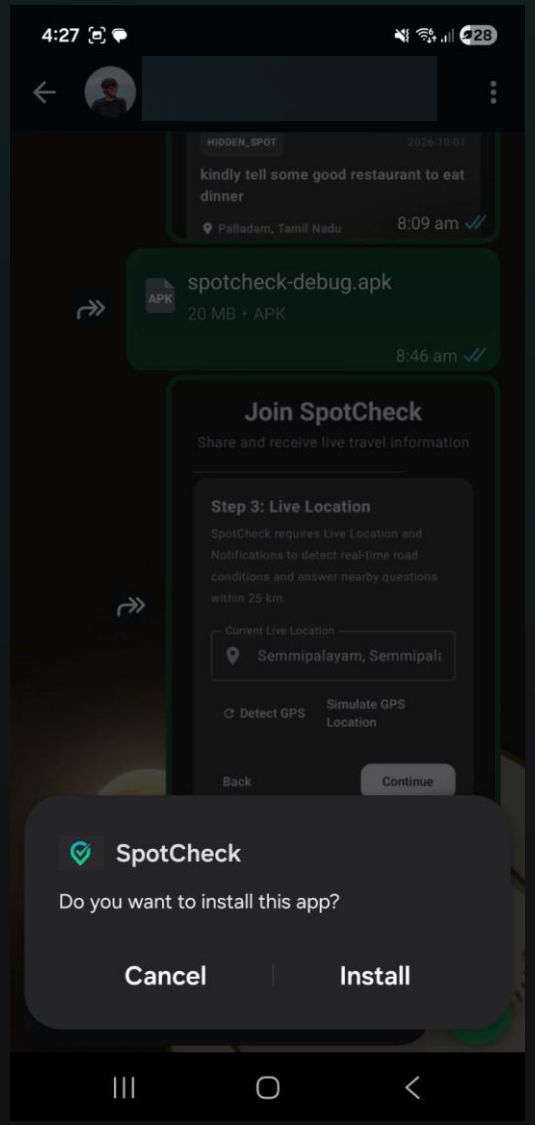
INSTALLATION & RUNTIME SECURITY

Part 1: Downloading & Installing the APK (~20 MB)

SpotCheck

- ✓ **Lightweight Distribution:** Delivered as a compact Android Package Kit (~20 MB) engineered for rapid installation across low-bandwidth networks.
- ✓ **Sandboxed Runtime:** Strict Android process isolation guarantees that private user data and session tokens remain inaccessible to other apps.
- ✓ **Step 1.1 Download Prompt:** Transparent browser and direct-link download with immediate 'Open file' execution flow.
- ✓ **Step 1.2 Package Confirmation:** Native system installer confirmation with zero unnecessary permission bloat.



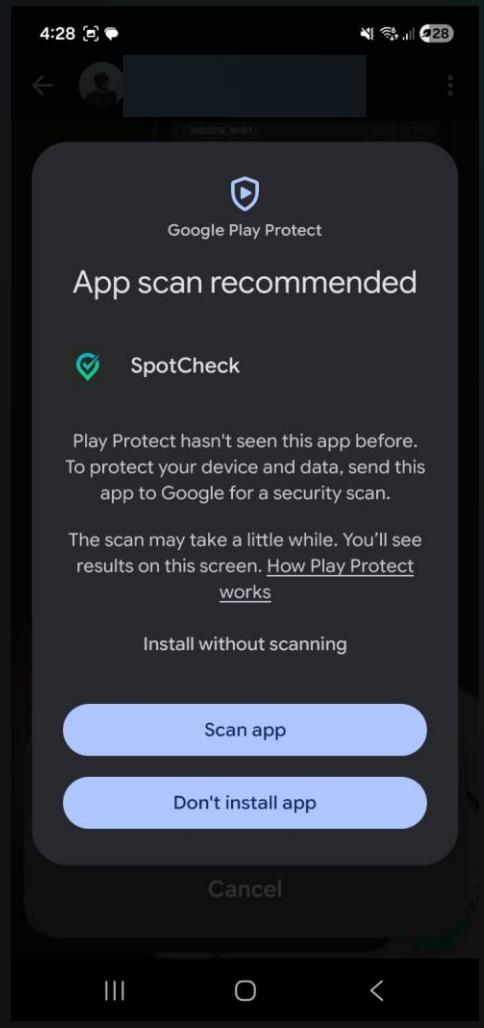


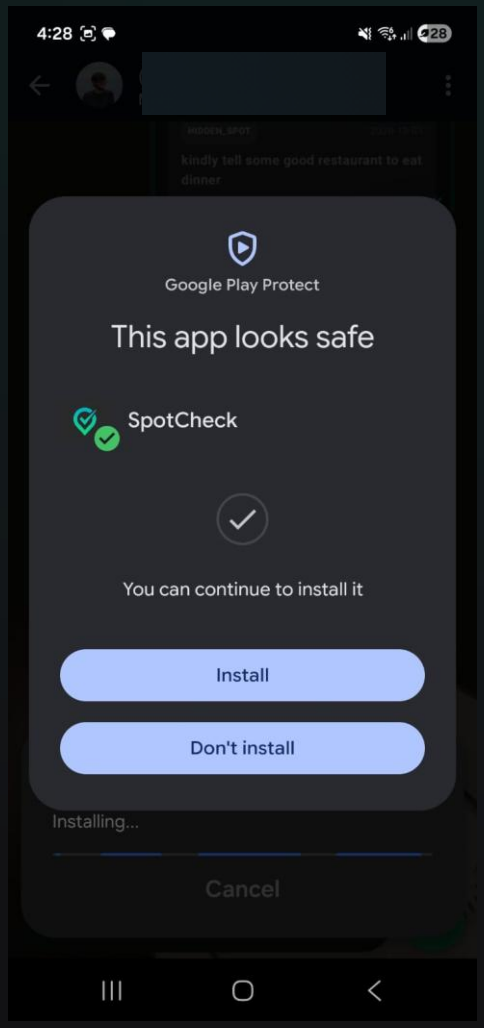
INSTALLATION & RUNTIME SECURITY

Part 1 (Cont.): Google Play Protect Verification

SpotCheck

- ✓ **Built-In Threat Inspection:** Google Play Protect examines bytecode and cryptographic signatures before execution.
- ✓ **Step 1.3 Scan Flexibility:** Allows one-tap 'Scan app' verification or immediate installation for certified enterprise deployments.
- ✓ **Step 1.4 Safe Verification:** Clean clearance confirms 100% clean application code, absence of malicious binaries, and verified integrity.
- ✓ **Launch-Ready Execution:** Instantly places the verified SpotCheck launcher icon on the Android home screen.



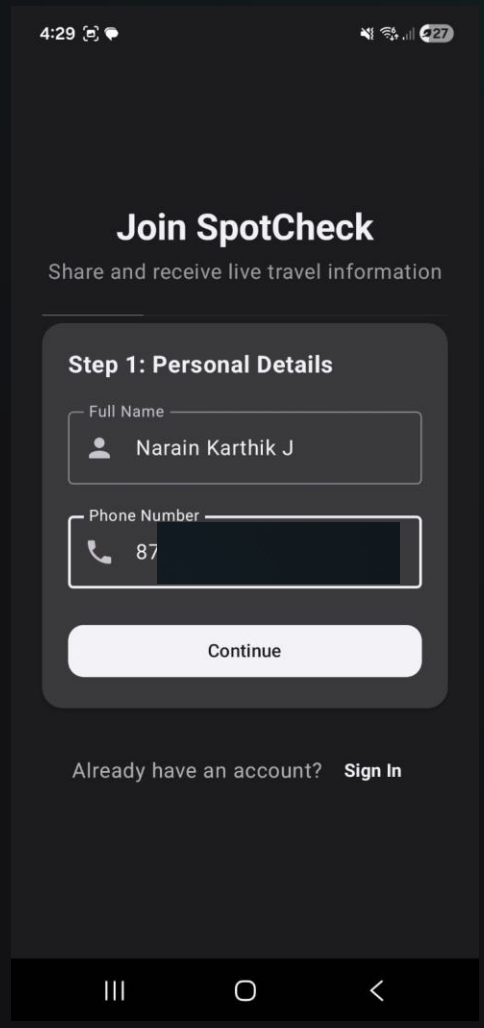


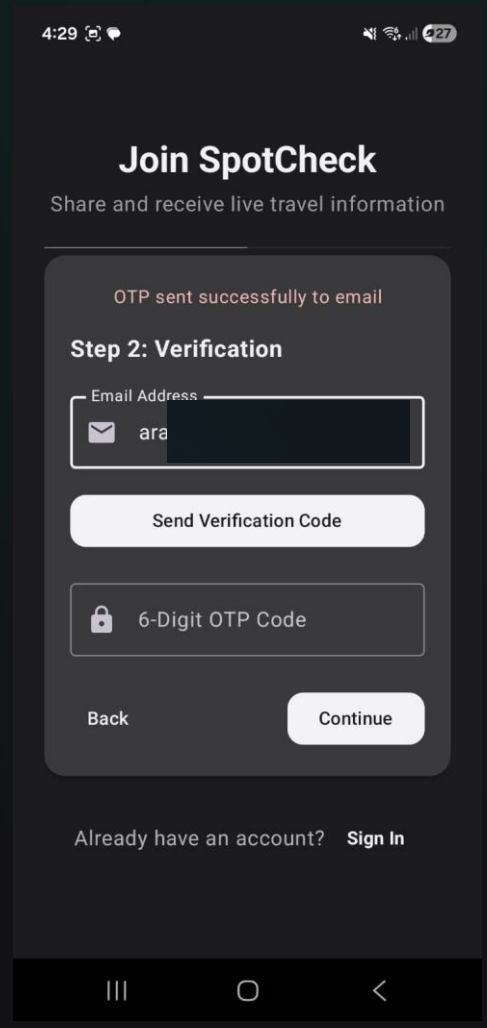
ONBOARDING & USER IDENTITY

Part 2: Account Creation — Personal Info & Bot-Proof OTP

SpotCheck

- ✓ **Step 2.1 Clean Sign In:** Streamlined login supporting existing accounts and instant one-tap registration for newcomers.
- ✓ **Step 2.2 Identification (Step 1/4):** Full legal name and active 10-digit mobile number input for transparent, accountable peer interactions.
- ✓ **Step 2.3 Bot-Proof Email OTP (Step 2/4):** Dispatches an instant 6-digit one-time passcode to verify authentic human ownership and stop automated bot spam.
- ✓ **Cryptographic Token Issuance:** Successful OTP verification prepares secure salted identity credentials in the backend.



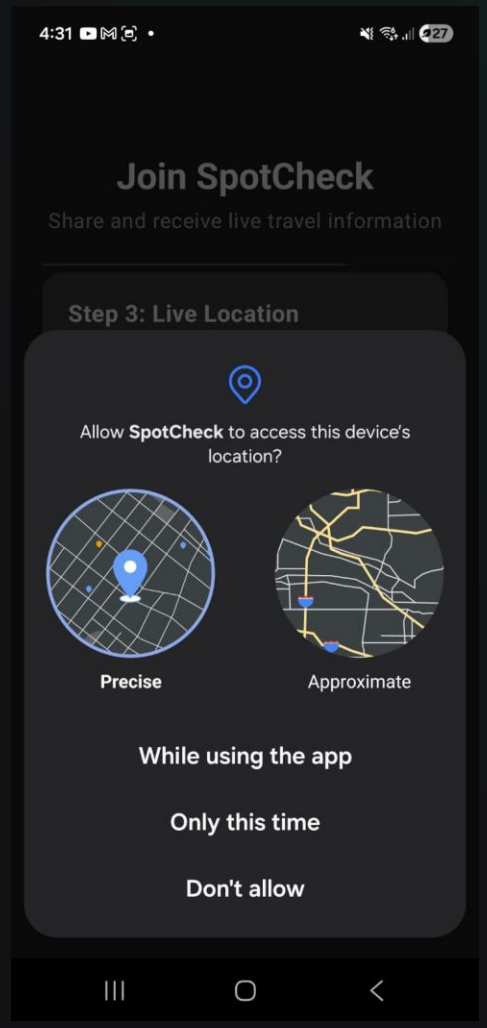


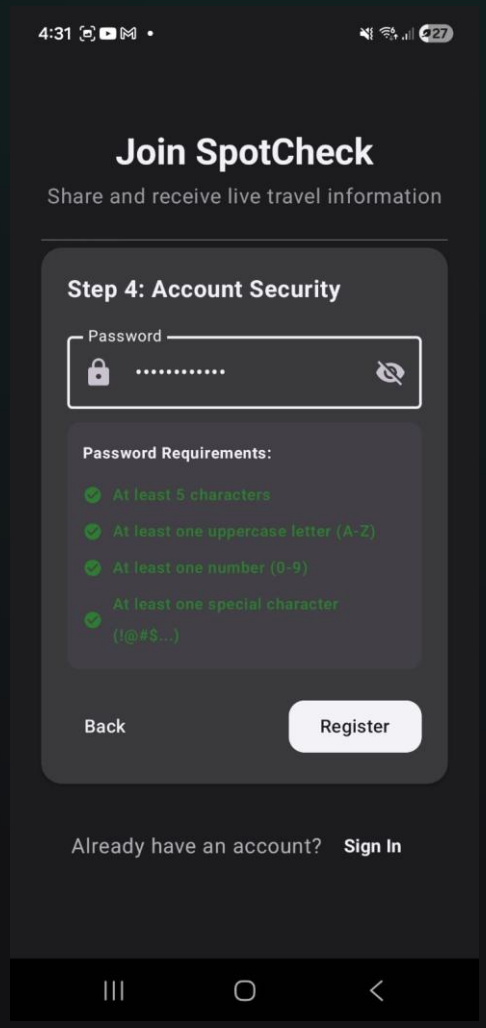
ONBOARDING & USER IDENTITY

Part 2 (Cont.): GPS Permissions & Password Standards

SpotCheck

- ✓ **Precise GPS Access:** Mandates high-accuracy satellite location permissions so users only receive questions and updates relevant to their immediate area.
- ✓ **Step 3/4 Live Location Confirmation:** Automated reverse-geocoding displays the detected locality (e.g., Semmipalayam) with one-tap refresh.
- ✓ **Step 4/4 Enforced Password Standards:** Real-time client-side validation requires 4 green checkmarks: 5+ characters, uppercase letter, number, and special character (!@#$).
- ✓ **Final Registration:** Creates verified traveler/native credentials and unlocks the real-time dashboard.



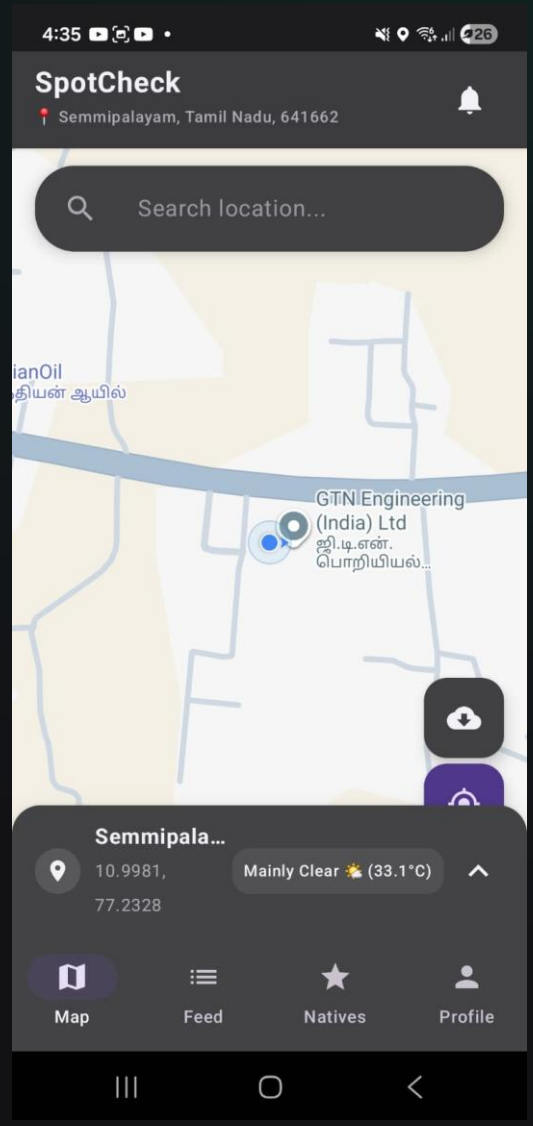


INTERACTIVE MAP & DASHBOARD

Part 3: Exploring the Live Map & Environmental Telemetry

SpotCheck

- ✓ **Live Satellite Fix:** Displays real-time positioning with an active blue marker and precise coordinate telemetry.
- ✓ **Universal Search Bar:** Direct search functionality to look up viewpoints, hill corridors, temples, and transit hubs before departing.
- ✓ **Live Weather Telemetry:** Embedded chip providing live meteorological updates for the active location (e.g., Mainly Clear 🌼 33.1°C).
- ✓ **Live Community Pulse:** Instant readout displaying the count of verified active locals currently nearby.
- ✓ **Unified Bottom Bar:** Seamless switching between Map, Community Feed, Leaderboard, and Profile settings.

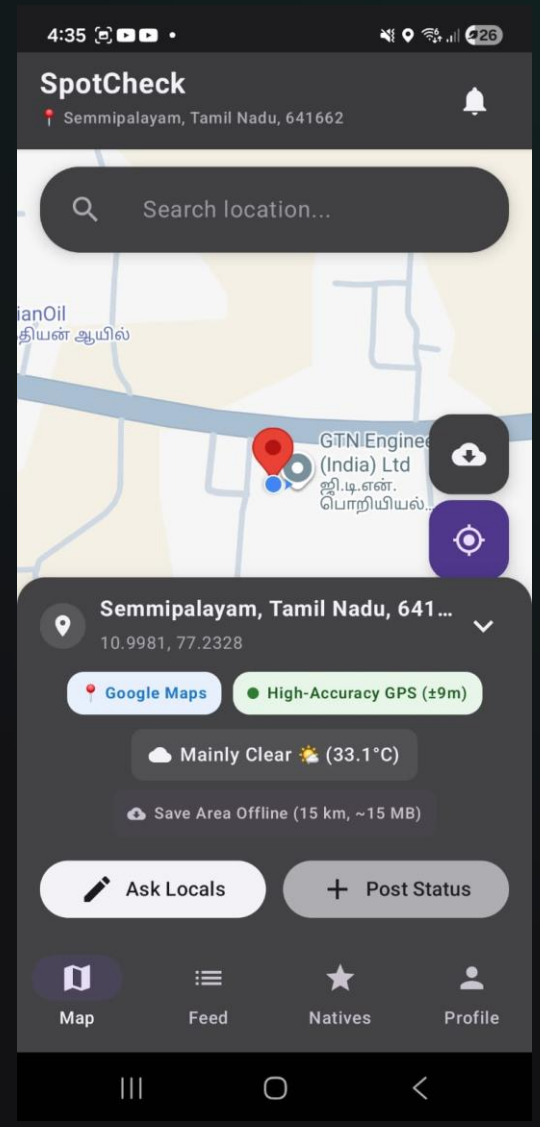


OFFLINE RESILIENCE

Part 3 (Cont.): Spot Action Sheet & 15 km Offline Cache

SpotCheck

- ✓ **Hardware Accuracy Readout:** Displays verified GPS precision fix (e.g., ±9 m) directly on the location inspection sheet.
- ✓ **External Navigation Link:** One-tap shortcut launches Google Maps for turn-by-turn routing directly from SpotCheck.
- ✓ **15 km Offline Area Cache (~15 MB):** Downloads vector map tiles and road topologies covering a 15 km radius to guarantee navigation across zero-reception mountain passes and ghats.
- ✓ **Action Hub:** Direct access to 'Ask Locals' for quick queries and 'Post Status' to share live ground updates.

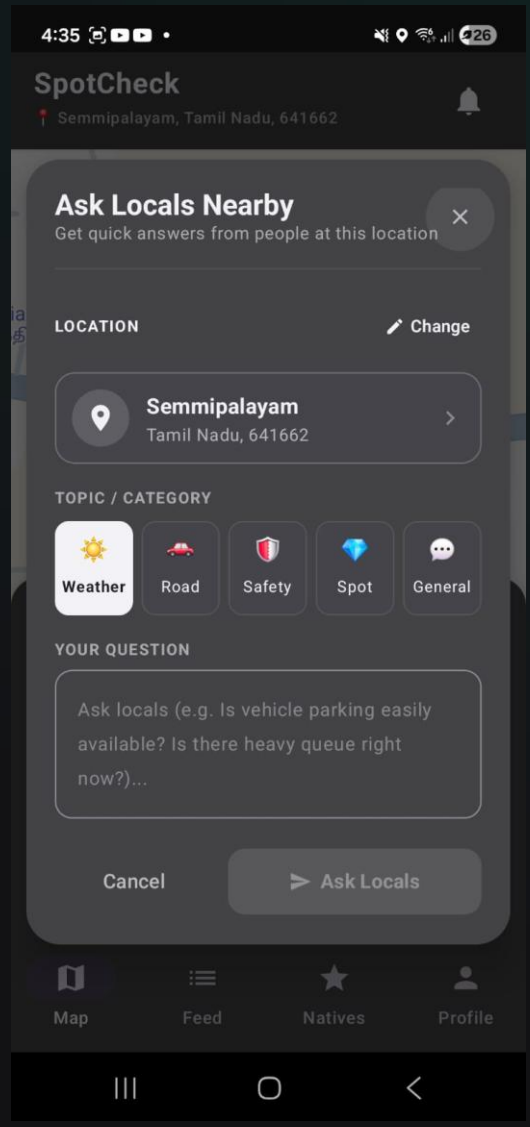


HYPER-LOCAL Q&A

Part 4: Asking Nearby Locals (25 km Geofence Dispatch)

SpotCheck

- ✓ **25 km Proximity Radius:** Inquiries are geofenced and dispatched exclusively to verified locals currently active within a 25 km radius.
- ✓ **5 Dedicated Query Categories:** 🌞 Weather (rain/fog) • 🚙 Road (landslides/repairs) • 🛡️ Safety (checkpoints/night travel) • 💎 Spot (crowds/hours) • 💬 General.
- ✓ **Instant Push Alerts:** Nearby residents receive real-time notifications alerting them to a traveler's immediate question.
- ✓ **Verified Distance Badging:** Incoming responses display the respondent's verified distance (e.g., 'Answered by Local 1.8 km away').

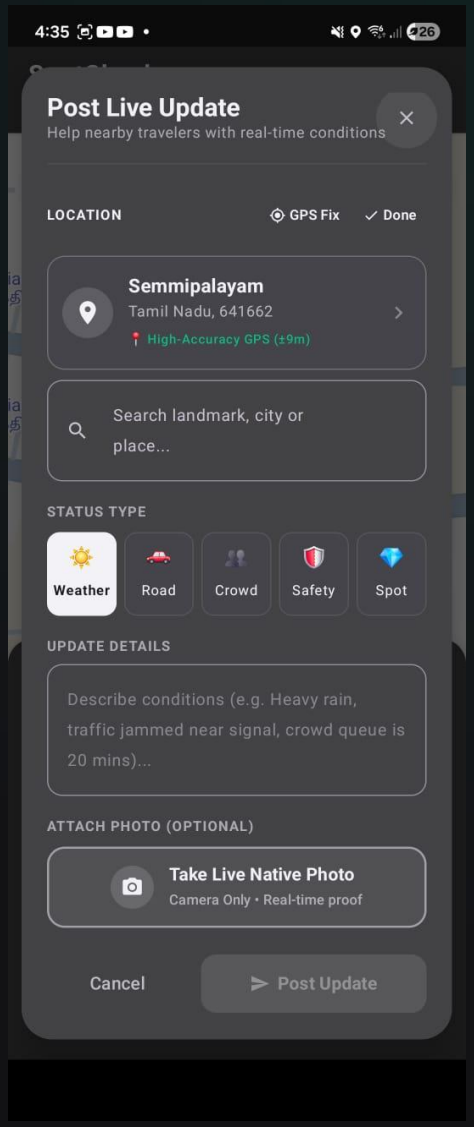


VERIFIED GROUND TRUTH

Part 5: Live Native Pulse & Zero-Tolerance Anti-Fraud

SpotCheck

- ✓ **Strict Elimination of Gallery Uploads:** File pickers and phone photo gallery uploads are disabled to prevent stale or downloaded internet photos.
- ✓ **Camera-Only Proof Enforcement:** Visual proof must be captured live through the hardware camera lens inside the SpotCheck app.
- ✓ **Satellite GPS Telemetry Matching:** Server-side algorithms compare photo capture metadata with active GPS coordinates to prevent spoofing.
- ✓ **Immediate Feed Publication:** Verified updates post instantly to both the live map and the community Native Pulse stream.

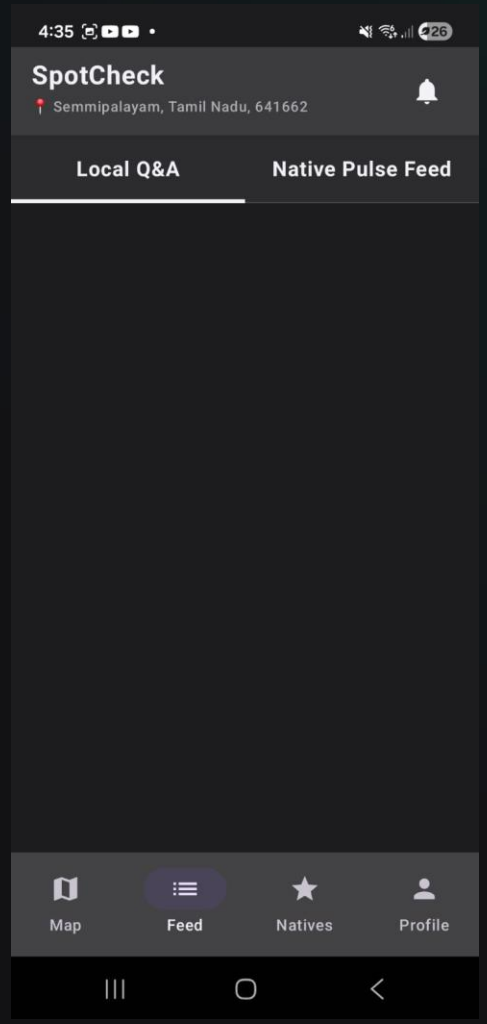


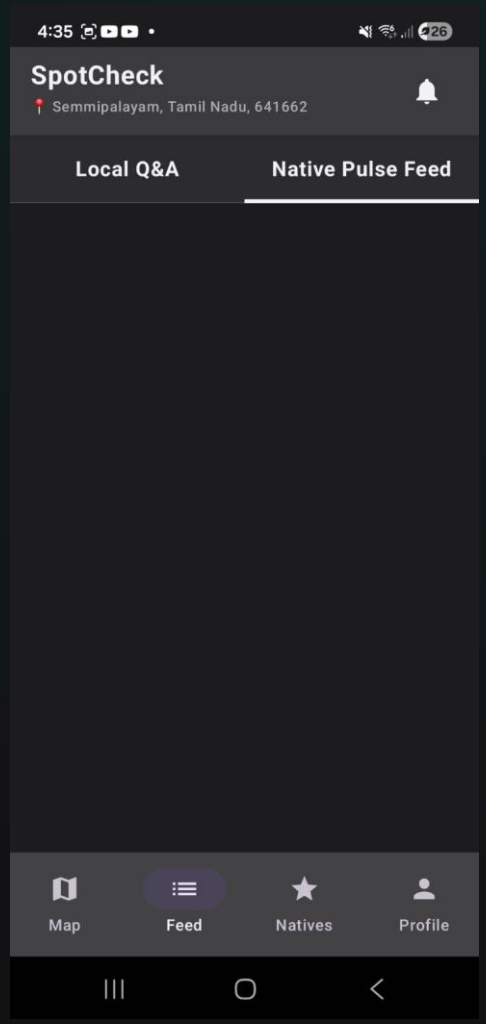
REAL-TIME COMMUNITY FEEDS

Part 6: Dual-Stream Community Feeds

SpotCheck

- ✓ **Stream 1: Local Q&A Feed:** Structured traveler questions organized by category; enables locals to discover questions and earn reward points.
- ✓ **Stream 2: Native Pulse Feed:** Chronological stream of live camera updates showing parking occupancy, weather shifts, and road conditions.
- ✓ **Automated Stale-Data Cleanup:** Posts older than 60–90 days are automatically archived to ensure zero clutter and 100% current relevance.
- ✓ **One-Tap Engagement:** Travelers can upvote verified answers and report inaccurate updates immediately.



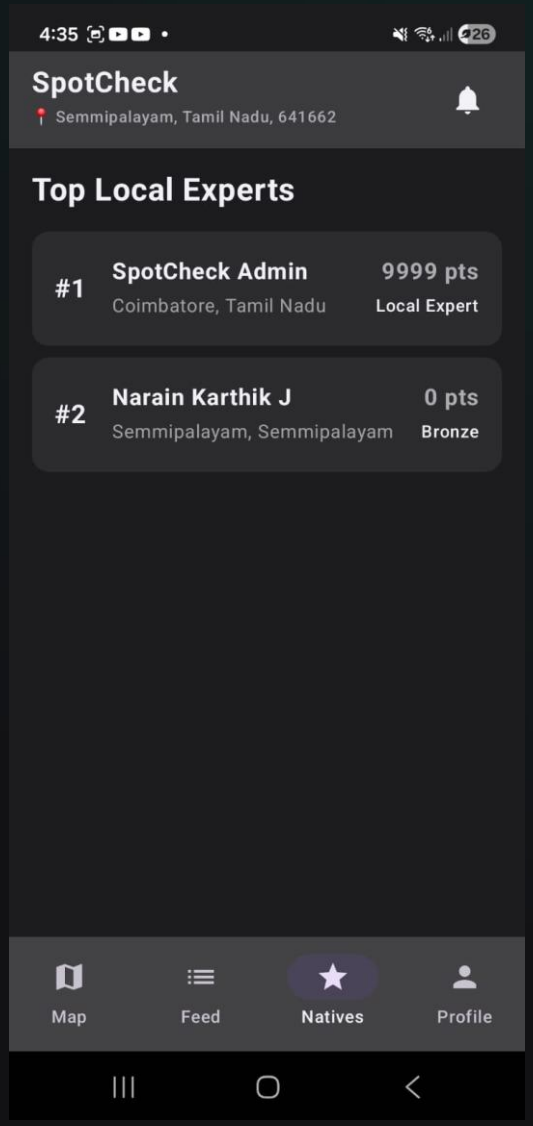


TRUST ECONOMY & REPUTATION

Part 7: Local Experts & Gamification Leaderboard

SpotCheck

- ✓ **Reward Credit Economy:** +5 Points for verified answers to traveler questions • +3 Points for authentic live camera pulse updates.
- ✓ **Progressive Reputation Tiers:** Bronze (0–100 pts) → Silver (101–500 pts) → Gold (501–1,499 pts) → Local Expert (1,500+ pts).
- ✓ **Misinformation Penalties:** Community flagging triggers automatic penalty deductions (-3 to -5 pts) and rank demotions for inaccurate reports.
- ✓ **Verified Native Shield:** Top-ranking local experts gain verified moderation badges and prominence across regional feeds.

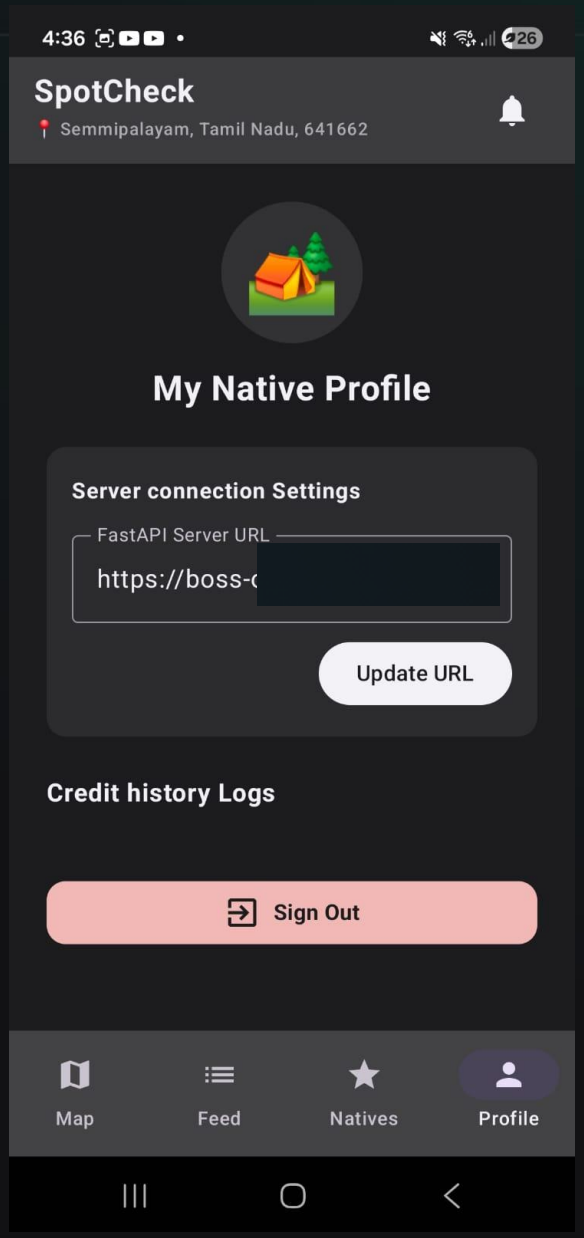


USER PROFILE & ADMINISTRATION

Part 8: Profile Management & Dynamic Server Settings

SpotCheck

- ✓ **Native Profile & Rank Display:** Shows user avatar, assigned reputation tier (e.g. Silver), total points earned, and total contributions.
- ✓ **Credit Transaction Ledger:** Full transparent audit history recording incoming reward credits and verification bonuses.
- ✓ **Dynamic FastAPI Server URL:** In-app configurable backend endpoint allowing instant switching between local staging, ngrok tunnels, and cloud production without app reinstalls.
- ✓ **Secure Sign Out:** Terminates active JWT session tokens and clears cached credentials cleanly.



UNDERLYING ENGINEERING

Part 9: Full Technical Stack & Architecture

SpotCheck

Android Client

100% Kotlin with Jetpack Compose declarative UI; Room DB for offline-first local persistence; Android WorkManager for periodic background sync.

FastAPI Backend Core

Python asynchronous backend providing sub-second query latency, Pydantic data validation, and automated OpenAPI documentation.

PostgreSQL Database

Stores relational profiles, questions, and answers with indexed spatial coordinate filtering for 25 km geofences.

Offline Vector Architecture

Pre-packaged vector tile caching system supporting 15 km continuous map rendering without cellular connectivity.



SpotCheck

Never travel on outdated expectations again. Travel on real-time ground truth.

- ✓ 25 km Hyper-Local Inquiries
- ✓ Anti-Fraud Live Camera Only
- ✓ 15 km Offline Area Cache
- ✓ Gamified Trust Economy

THINAIRA tech

"Nilam Kaathu, Neri Nindru, Thisai Kaattum Payanam — Thinai!"